
EECS 16A Designing Information Devices and Systems I

Discussion 5B

1. Mechanical Eigenvalues and Eigenvectors

In each part, find the eigenvalues of the matrix and the associated eigenspaces. State if the inverse of the matrix exists.

(a) $\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$

(b) $\mathbf{A} = \begin{bmatrix} -2 & 4 \\ -4 & 8 \end{bmatrix}$

(c) $\mathbf{A} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$

(d) $\mathbf{M} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$

(e) **(PRACTICE)** $\mathbf{M} = \begin{bmatrix} 1 & 0 \\ 0 & 9 \end{bmatrix}$

2. Steady and Unsteady States

(a) You're given the matrix $\mathbf{M}$:

$$\mathbf{M} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ 0 & 1 & -2 \\ 0 & 0 & 2 \end{bmatrix}$$

Which generates the next state of a physical system from its previous state: $\vec{x}[k+1] = \mathbf{M}\vec{x}[k]$. ($\vec{x}$ could describe people or water, for example) Find the eigenspaces associated with the following eigenvalues:

- i. $\text{span}(\vec{v}_1)$, associated with $\lambda_1 = 1$
- ii. $\text{span}(\vec{v}_2)$, associated with $\lambda_2 = 2$
- iii. $\text{span}(\vec{v}_3)$, associated with $\lambda_3 = \frac{1}{2}$

(b) Define $\vec{x} = \alpha\vec{v}_1 + \beta\vec{v}_2 + \gamma\vec{v}_3$, a linear combination of the eigenvectors. For each of the cases in the table, determine if

$$\lim_{n \rightarrow \infty} \mathbf{M}^n \vec{x}$$

converges. If it does, what does it converge to?

α	β	γ	Converges?	$\lim_{n \rightarrow \infty} \mathbf{M}^n \vec{x}$
0	0	$\neq 0$		
0	$\neq 0$	0		
0	$\neq 0$	$\neq 0$		
$\neq 0$	0	0		
$\neq 0$	0	$\neq 0$		
$\neq 0$	$\neq 0$	0		
$\neq 0$	$\neq 0$	$\neq 0$		